Characteristics of a Data Structure

- **Correctness** − Data structure implementation should implement its interface correctly.
- **Time Complexity** − Running time or the execution time of operations of data structure must be as small as possible.
- **Space Complexity** − Memory usage of a data structure operation should be as little as possible.

Execution Time Cases

There are three cases which are usually used to compare various data structure's execution time in a relative manner.

- **Worst Case** − This is the scenario where a particular data structure operation takes maximum time it can take. If an operation's worst case time is ƒ(n) then this operation will not take more than ƒ(n) time where ƒ(n) represents function of n.
- **Average Case** − This is the scenario depicting the average execution time of an operation of a data structure. If an operation takes ƒ(n) time in execution, then m operations will take mƒ(n) time.
- **Best Case** − This is the scenario depicting the least possible execution time of an operation of a data structure. If an operation takes ƒ(n) time in execution, then the actual operation may take time as the random number which would be maximum as ƒ(n).

Algorithm

is a step-by-step procedure, which defines a set of instructions to be executed in a certain order to get the desired output. Algorithms are generally created independent of underlying languages, i.e. an algorithm can be implemented in more than one programming language.

From the data structure point of view, following are some important categories of algorithms

- **Search** − Algorithm to search an item in a data structure.
- **Sort** − Algorithm to sort items in a certain order.
- **Insert** − Algorithm to insert item in a data structure.
- **Update** − Algorithm to update an existing item in a data structure.
- **Delete** − Algorithm to delete an existing item from a data structure.

Characteristics of an Algorithm

Not all procedures can be called an algorithm. An algorithm should have the following characteristics −

- **Unambiguous** − Algorithm should be clear and unambiguous. Each of its steps (or phases), and their inputs/outputs should be clear and must lead to only one meaning.
- **Input** − An algorithm should have 0 or more well-defined inputs.
- **Output** − An algorithm should have 1 or more well-defined outputs, and should match the desired output.
- **Finiteness** − Algorithms must terminate after a finite number of steps.
- **Feasibility** − Should be feasible with the available resources.
- **Independent** − An algorithm should have step-by-step directions, which should be independent of any programming code.

Algorithm Complexity

Suppose **X** is an algorithm and **n** is the size of input data, the time and space used by the algorithm X are the two main factors, which decide the efficiency of X.

- **Time Factor** − Time is measured by counting the number of key operations such as comparisons in the sorting algorithm.
- **Space Factor** − Space is measured by counting the maximum memory space required by the algorithm.

The complexity of an algorithm **f(n)** gives the running time and/or the storage space required by the algorithm in terms of **n** as the size of input data.

Space Complexity

Space complexity of an algorithm represents the amount of memory space required by the algorithm in its life cycle. The space required by an algorithm is equal to the sum of the following two components −

- A fixed part that is a space required to store certain data and variables, that are independent of the size of the problem. For example, simple variables and constants used, program size, etc.
- A variable part is a space required by variables, whose size depends on the size of the problem. For example, dynamic memory allocation, recursion stack space, etc.

Time Complexity

Time complexity of an algorithm represents the amount of time required by the algorithm to run to completion. Time requirements can be defined as a numerical function T(n), where T(n) can be measured as the number of steps, provided each step consumes constant time.

For example, addition of two n-bit integers takes n steps. Consequently, the total computational time is $T(n) = c * n$, where c is the time taken for the addition of two bits. Here, we observe that T(n) grows linearly as the input size increases.

Asymptotic Analysis

Asymptotic analysis of an algorithm refers to defining the mathematical foundation/framing of its run-time performance. Using asymptotic analysis, we can very well conclude the best case, average case, and worst case scenario of an algorithm.

Asymptotic analysis is input bound i.e., if there's no input to the algorithm, it is concluded to work in a constant time. Other than the "input" all other factors are considered constant.

Asymptotic analysis refers to computing the running time of any operation in mathematical units of computation. For example, the running time of one operation is computed as f(n) and may be for another operation it is computed as $g(n^2)$. This means the first operation running time will increase linearly with the increase in **n** and the running time of the second operation will increase exponentially when **n** increases. Similarly, the running time of both operations will be nearly the same if **n** is significantly small.

Usually, the time required by an algorithm falls under three types

- **Best Case** − Minimum time required for program execution.
- **Average Case** − Average time required for program execution.
- **Worst Case** − Maximum time required for program execution.

Asymptotic Notations

Execution time of an algorithm depends on the instruction set, processor speed, disk I/O speed, etc. Hence, we estimate the efficiency of an algorithm asymptotically.

Time function of an algorithm is represented by T(n), where n is the input size.

Different types of asymptotic notations are used to represent the complexity of an algorithm. Following asymptotic notations are used to calculate the running time complexity of an algorithm.

O − Big Oh Notation

Ω − Big omega Notation

θ − Big theta Notation

o − Little Oh Notation

ω − Little omega Notation

Big Oh, O: Asymptotic Upper Bound

The notation (n) is the formal way to express the upper bound of an algorithm's running time. is the most commonly used notation. It measures the **worst case time complexity** or the longest amount of time an algorithm can possibly take to complete.

Big Omega, Ω: Asymptotic Lower Bound

The notation Ω(n) is the formal way to express the lower bound of an algorithm's running time. It measures the **best case time complexity** or the best amount of time an algorithm can possibly take to complete.

Theta, θ: Asymptotic Tight Bound

The notation (n) is the formal way to express both the lower bound and the upper bound of an algorithm's running time. Some may confuse the theta notation as the average case time complexity; while big theta notation could be *almost* accurately used to describe the average case, other notations could be used as well.

Common Asymptotic Notations

Following is a list of some common asymptotic notations −

constant	–	$O(1)$
logarithmic	–	$O(\log n)$
linear	–	$O(n)$
n log n	–	$O(n \log n)$
quadratic	–	$O(n^2)$
cubic	–	$O(n^3)$
polynomial	–	$n^{O(1)}$
exponential	–	$2^{O(n)}$

Data Type

Data type is a way to classify various types of data such as integer, string, etc. which determines the values that can be used with the corresponding type of data, the type of operations that can be performed on the corresponding type of data. There are two data types

Built-in Data Type

Derived Data Type

Built-in Data Type

Those data types for which a language has built-in support are known as Built-in Data types. For example, most of the languages provide the following built-in data types.

Integers

Boolean (true, false)

Floating (Decimal numbers)

Character and Strings

Derived Data Type

Those data types which are implementation independent as they can be implemented in one or the other way are known as derived data types. These data types are normally built by the combination of primary or built-in data types and associated operations on them. For example

List

Array

Stack

Queue

Basic Operations

The data in the data structures are processed by certain operations. The particular data structure chosen largely depends on the frequency of the operation that needs to be performed on the data structure.

Traversing

Searching

Insertion

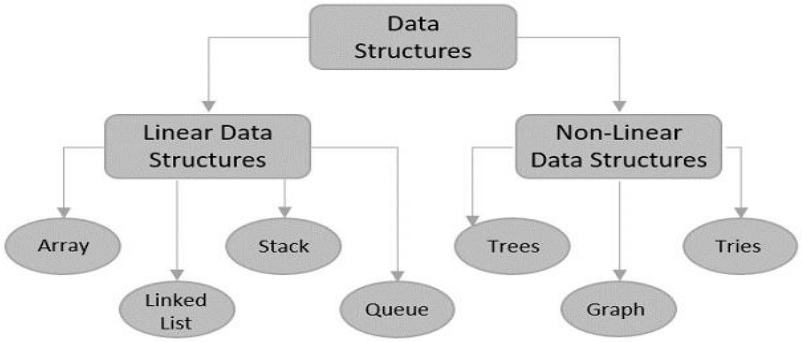
Deletion

Sorting

Merging

Data structures are introduced in order to store, organize and manipulate data in programming languages. They are designed in a way that makes accessing and processing of the data a little easier and simpler. These data structures are not confined to one particular programming language; they are just pieces of code that structure data in the memory.

Data types are often confused as a type of data structures, but it is not precisely correct even though they are referred to as Abstract Data Types. Data types represent the nature of the data while data structures are just a collection of similar or different data types in one.



There are usually just two types of data structures −

- Linear
- Non-Linear

Linear Data Structures

The data is stored in linear data structures sequentially. These are rudimentary structures since the elements are stored one after the other without applying any mathematical operations.

Linear data structures are usually easy to implement but since the memory allocation might become complicated, time and space complexities increase. Few examples of linear data structures include

- Arrays
- Linked Lists
- Stacks
- Queues

Based on the data storage methods, these linear data structures are divided into two sub-types. They are − **static** and **dynamic** data structures.

Static Linear Data Structures

In Static Linear Data Structures, the memory allocation is not scalable. Once the entire memory is used, no more space can be retrieved to store more data. Hence, the memory is required to be reserved based on the size of the program. This will also act as a drawback since reserving more memory than required can cause a wastage of memory blocks.

The best example for static linear data structures is an array.

Dynamic Linear Data Structures

In Dynamic linear data structures, the memory allocation can be done dynamically when required. These data structures are efficient considering the space complexity of the program.

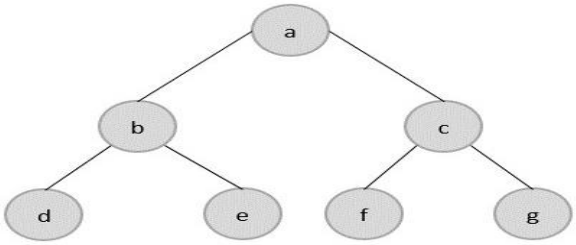
Few examples of dynamic linear data structures include: linked lists, stacks and queues.

Non-Linear Data Structures

Non-Linear data structures store the data in the form of a hierarchy. Therefore, in contrast to the linear data structures, the data can be found in multiple levels and are difficult to traverse through.

types of non-linear data structures are −

I. Graphs
II. Trees
III. Tries
IV. Maps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